

1(a) Thermal voltage of a diode at 25°C .

$$T_c = 25^{\circ}\text{C}$$

$$T_k = T_c + 273 = 25 + 273 = 298\text{K}.$$

$$\text{Boltzmann constant, } K = 1.38 \times 10^{-23} \text{ J/K}$$

$$\text{charge of an electron, } q = 1.6 \times 10^{-19} \text{ C}$$

Thermal voltage:

$$V_T = \frac{KT}{q}$$

$$V_T = \frac{1.38 \times 10^{-23} \times 298}{1.6 \times 10^{-19}}$$

$$V_T \approx 0.0257 \text{ V}$$

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$$\approx \underline{\underline{26 \text{ mV}}}.$$

$$\text{b) } I_s = 40 \text{ nA} = 40 \times 10^{-9} \text{ A}$$

$$n = 2$$

$$V_D = 0.5 \text{ V}$$

$$V_T = 0.026 \text{ V}$$

$$I_D = I_s \left(e^{\frac{V_D}{nV_T}} - 1 \right)$$

$$I_D = 40 \times 10^{-9} \left(e^{\frac{0.5}{2 \times 0.026}} - 1 \right)$$

$$\frac{0.5}{0.052} \approx 9.62$$

$$I_D = 40 \times 10^{-9} (e^{9.62})$$

$$I_D = 40 \times 10^{-9} \times 1.5 \times 10^4$$

$$I_D \approx 6.0 \times 10^{-4} \text{ A}$$

$$I_D \approx \underline{\underline{0.6 \text{ mA}}}.$$